Functional Analysis II - MATH7321

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Chapter 1

Banach Algebras

14/01/2025 Let $\mathcal{H}$ be a separable Hilbert space, and $T \in \mathcal{K}(\mathcal{H})$ be self adjoint, and X be the orthonormal basis for $\mathcal{H}$ consisting of the eigenvectors of T. Then $\mathcal{H} \cong \ell^2(X)$ and then T can be identified with a multiplication operator M_f with $f \in \ell^\infty(X)$ such that $f(x) = \lambda_x$, where λ_x is the eigenvalue for the eigenvector, $x \in X$.

If $T \in B(\mathcal{H})$ is normal, then $\operatorname{span}\{T^m T^{*n} \ : \ m, n \in \mathbb{N}\}$ is an involutive algebra.

1.1 Preliminaries

Definition 1.1.1. An algebra over $\mathbb{C}$(or $\mathbb{R}$) is a vector space over $\mathbb{C}$(or $\mathbb{R}$) with a product that is associative and distributive.

product is continuous A normed algebra is an algebra $\mathcal{A}$ equipped with a norm such that $\|ab\| \leq \|a\|\|b\|$ for all $a, b \in \mathcal{A}$.

If the norm is complete in a normed algebra, we call it a Banach algebra.

An algebra is unital if has a neutral element 1 with respect to the multiplication. It is called commutative if $ab = ba$ for all $a, b \in \mathcal{A}$

An involution on an algebra is a map $* : \mathcal{A} \to \mathcal{A}$ such that

(1) $(\alpha a + b)^* = \overline{\alpha} a^* + b^*$

(2) $(ab)^* = b^* a^*$

(3) $(a^*)^* = a$

for all $a, b \in \mathcal{A}$ and $\alpha \in \mathbb{C}$

A normed $*$-algebra is a normed algebra with an involution $*$ such that $\|a^*\| = \|a\|$.

Definition 1.1.2. Let $\mathcal{A}$ be a unital Banach algebra. An element $a \in \mathcal{A}$ is invertible if there exists $b \in \mathcal{A}$ such that $ab = ba = 1$.

The spectrum of a is the set

$$\sigma(a) = \{\lambda \in \mathbb{C} \ : \ \lambda 1 - a \text{ is not invertible } \}$$

Lemma 1.1.1. *In any unital Banach algebra, the set of invertible elements is open.*

Proof. If $\|x\| < 1$, then

$$\frac{1}{1-x} = \sum_{n \in \mathbb{N}} x^n$$

or equivalently if $\|1 - x\| < 1$, then

$$\frac{1}{x} = \sum_{n \in \mathbb{N}} (1-x)^n$$

Now let x_0 be invertible in the unital Banach algebra and let $x \in B(x_0, \frac{1}{\|x_0^{-1}\|})$. Then $\|xx_0^{-1} - 1\| \leq \|x - x_0\|\|x_0^{-1}\| < 1$ and similarly $\|x_0^{-1}x - 1\| < 1$. This shows that $xx_0^{-1}, x_0^{-1}x$ are invertible with inverses (assumed) y, z respectively. Thus $xx_0^{-1}y = 1 = zx_0^{-1}x$. verify. □

Definition 1.1.3. The spectral radius of $a \in \mathcal{A}$ is defined to be

$$r(a) = \sup\{|\lambda| \ : \ \lambda \in \sigma(a)\}$$

Lemma 1.1.2. *For all $a \in \mathcal{A}$,*

$$r(a) \leq \|a\|$$

Theorem 1.1.1. *Let $\mathcal{A}$ be a unital Banach algebra and $a \in \mathcal{A}$. Then $\sigma(a)$ is a non-empty compact subset of $\mathbb{C}$.*

Proof. proof incoming for non-empty. See last semester notes for proof of compactness. □

Lemma 1.1.3. *Let X be a normed space. Then there exist a compact set K and an isometric linear map $X \to C(K)$.*

Proof. embed X to X^{**}, with K being the weak * closed unit ball of X^* □

Definition 1.1.4. Let $\mathcal{A}$ be a commutative Banach algebra. The spectrum of $\mathcal{A}$ is the set $\mathrm{sp}(\mathcal{A}) = \{\tau \in \mathcal{A}^* \ : \ \tau(ab) = \tau(a)\tau(b), a, b \in \mathcal{A}\}$.

Lemma 1.1.4. *For every $\tau \in sp(\mathcal{A})$,*

1. $\tau(1) = 1$
2. $\|\tau\| = 1$
3. *sp*$(\mathcal{A})$ *is a weak * closed subset of* $\mathcal{A}^*$ *(Hence compact)*

Proof. (1) Since $\tau(a) = \tau(a \cdot \mathbf{1}) = \tau(a)\tau(\mathbf{1})$ for all $a \in \mathcal{A}$, it is immediate that $\tau(\mathbf{1}) = 1 \in \mathbb{C}$.

(2) Since $\tau(\mathbf{1}) = 1$, it is clear that $\|\tau\| \geq 1$. The other inequality follows from the fact that $\tau(a) \in \sigma(a)$ (see that $a - \tau(a)\mathbf{1} \in \mathrm{Ker}(\tau)$) and thus $|\tau(a)| \leq r(a) \leq \|a\|$.

(3) Let $\tau, \phi \in \mathrm{sp}(A)$. Then for all $\hat{a} \in \hat{\mathcal{A}}$ (image of $\mathcal{A}$ in $\mathcal{A}^{**}$ under the natural inclusion), we have

$$|\hat{a}(\tau - \phi)| = |\hat{a}(\tau) - \hat{a}(\phi)| = |\tau(a) - \phi(a)| = |(\tau - \phi)(a)| \leq \|a\||\tau - \phi|$$

Thus if τ_α is any weak * Cauchy net, $\tau_\alpha(a)$ converges for all $a \in \mathcal{A}$. Let $\tau(a) = \lim_\alpha \tau_\alpha(a)$. Then all we have left is to prove that $\tau \in \mathrm{sp}(\mathcal{A})$. But since $\tau_(\alpha) \to \tau(a)$ and $\tau_\alpha(b) \to \tau(b)$, by the algebra of limits in $\mathbb{C}$, we get $\tau(ab) = \tau(a)\tau(b)$ proving $\tau \in \mathrm{sp}(\mathcal{A})$.

Thus we see that $\mathrm{sp}(\mathcal{A})$ is a weak * closed subspace of the unit ball of $\mathcal{A}^*$ which is closed by the Banach Alaoglu theorem.

□

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Definition 1.1.5. An ideal $I \subsetneq \mathcal{A}$ is called maximal if for any ideal $I \subset J$, then either $J = I$ or $J = \mathcal{A}$

Proposition 1.1.1. *Recall that R/I is a field if and only if I is a maximal ideal of the ring R.*

Remark 1.1.1. While we consider ideals of an algebra, since the algebra has a vector space structure, we demand the ideal to be subspace with respect to the underlying linear operations

Lemma 1.1.5. *If $\mathcal{A}$ is a (unital) Banach algebra, and I is a closed ideal, then $\mathcal{A}/I$ is a (unital) Banach algebra.*

Proof. Since I is an ideal, the ring structure of A/I is well defined. Moreover since $\mathcal{A}$ is a Banach space and I is a closed subspace, we know that $\mathcal{A}/I$ is a Banach space. Thus, we just need to verify the norm inequality

$$\|ab + I\| \leq \|a + I\|\|b + I\|$$

By the definition of the quotient norm for any $\varepsilon > 0$, there exist a $i_a, i_b \in I$ such that

$$\|a + i_a\| \leq \|a + I\| + \varepsilon, \quad \|b + i_b\| \leq \|b + I\| + \varepsilon$$

Then,

$$\begin{aligned}
\|ab + I\| &\leq \|ab + i_a b + i_b a + i_a i b\| \\
&= \|(a + i_a)(b + i_b)\| \\
&\leq \|a + i_a\|\|b + i_b\| \\
&\leq (\|a + I\| + \varepsilon)(\|b + I\| + \varepsilon) \\
&\leq \|a + I\|\|b + I\| + \varepsilon\|b + I\| + \varepsilon\|a + I\| + \varepsilon^2
\end{aligned}$$

Since ε was chosen arbitrarily, this gives our result. □

Lemma 1.1.6. *In a unital Banach algebra, every maximal ideal is closed.*

Proof. Take a maximal ideal, take its closure, then it must be either the ideal itself or the whole of the algebra. If it contains the whole of the algebra, then the unital element must be there. Then the original ideal must contain invertible elements by the openness of the set of invertible elements. This will make the original ideal, the whole of the algebra, which is a contradiction. □

Lemma 1.1.7. *If $\mathcal{A}$ is a Banach algebra that is a division ring (If every non-zero element has an inverse), then $\mathcal{A} = \mathbb{C}$.*

Proof. Let $0 \neq a \in \mathcal{A}$. Let $\lambda \in \sigma(a)$. Then $\lambda 1 - a$ is not invertible. Hence $\lambda 1 - a = 0$. So $a = \lambda 1$. Hence $\mathcal{A} = \mathbb{C}$. □

Corollary 1.1.1.1. *Let $\mathcal{A}$ be a unital commutative Banach algebra, and I be a maximal ideal. Then $\mathcal{A}/I = \mathbb{C}$.*

Lemma 1.1.8. *For every $a \in \mathcal{A}$, a commutative unital Banach algebra,*

$$\sigma(a) = \{\tau(a) \ : \ \tau \in sp(\mathcal{A})\}$$

Proof. Let $\tau \in \mathrm{sp}(\mathcal{A})$, then $\tau(\tau(a)1 - a) = 0$ and therefore $\tau(a)1 - a \in \mathrm{Ker}(\tau)$, hence it is not invertible. Hence $\tau(a) \in \sigma(a)$.

Conversely, let $\lambda \in \sigma(a)$, then $\lambda 1 - a$ is not invertible, thus the ideal $I = \langle \lambda 1 - a \rangle$ is a proper ideal, since $r(\lambda 1 - a)$ will not be invertible for any $r \in \mathcal{A}$. So $1 \notin \langle \lambda 1 - a \rangle$. By Zorn's lemma, $\langle \lambda 1 - a \rangle$ is contained in a maximal ideal I_λ.

Define $\tau : \mathcal{A} \to \mathbb{C} = \mathcal{A}/I_\lambda := x \mapsto x + I_\lambda$. Then $\tau \in \mathrm{sp}(\mathcal{A})$, and $\tau(a) = a + I_\lambda = \lambda + I_\lambda$ since $\lambda 1 - a \in I_\lambda$. Now by the identification of A/I_λ with $\mathbb{C}$ as in Lemma 1.1.7, we see that $\tau(a) = \lambda$. □

Definition 1.1.6. Let $\mathcal{A}$ be a commutative Banach algebra. Define

$$\Phi : \mathcal{A} \to C(\mathrm{sp}(\mathcal{A})) := \Phi(a)(\tau) = \tau(a)$$

for all $a \in \mathcal{A}, \tau \in \mathrm{sp}(\mathcal{A})$. The map Φ is called the **Gelfand transform**.

Theorem 1.1.2. *Φ is a contractive algebra homomorphism with $\|\Phi(a)\| = r(a)$.*

Proof. That Φ is contractive follows easily from

$$\|\Phi(a)(\tau)\| = \|\tau(a)\| \leq \|a\|$$

since τ is a contraction as proved in Lemma 1.1.4. Linearity and multiplicativity of Φ follows form the fact that every element $\tau \in \mathrm{sp}(\mathcal{A})$ is linear and multiplicative on $\mathcal{A}$. □

Remark 1.1.2 (Maximal ideals of $\mathcal{A}$ and $\mathrm{sp}(\mathcal{A})$). Let $\mathcal{A}$ be unital commutative Banach algebra. Let $\tau \in \mathrm{sp}(\mathcal{A})$, then $\mathrm{Ker}(\tau)$ is a closed ideal of $\mathcal{A}$, and $A/\mathrm{Ker}(\tau) \cong \mathbb{C}$ by the first isomorphism theorem. So $\mathrm{Ker}(\tau)$ is a maximal ideal. The converse of this is also true. Natural map to the quotient space of a maximal ideal (which is now a filed isomorphic to $\mathbb{C}$) gives an element of the $\mathrm{sp}(\mathcal{A})$. Hence $\mathrm{sp}(\mathcal{A})$ can be identified with the maximal ideals of $\mathcal{A}$.

Remark 1.1.3. Suppose $\tau, \tau' \in \mathrm{sp}(\mathcal{A})$, with $\mathrm{Ker}(\tau) = \mathrm{Ker}(\tau')$. Let $a \in \mathcal{A}$, then $\tau(a)1 - a \in \mathrm{Ker}(\tau) = \mathrm{Ker}(\tau')$ implies $\tau(a) = \tau'(a)$ for all $a \in \mathcal{A}$.

Remark 1.1.4. Combining both of the above, we see that $\mathrm{Ker}(\Phi)$ is the intersection of all maximal ideals of $\mathcal{A}$, that is the radical of $\mathcal{A}$.

Theorem 1.1.3. *Let $\mathcal{A}$ be a Banach algebra. Then $\forall a \in \mathcal{A}$, we have*

$$r(a) = \lim_{n\to\infty} \|a^n\|^{1/n}$$

Proof. verify □

Corollary 1.1.3.1. *If $\|a^2\| = \|a\|^2$, then $r(a) = \|a\|$ and the Gelfand transform will be halal.*

Example 1.1.1. Let $T \in B(\mathcal{H})$ be self-adjoint. Let $\mathcal{A} = \overline{\mathrm{span}}\{T^n \ : \ n \in \mathbb{N} \cup 0\}$. Then $\mathcal{A}$ is a unital Banach algebra. Moreover we have

$$\|T^2\| = \|T\|^2$$

by the self adjointness of T. Thus the Gelfand transform Φ is isometric on $\mathbb{R}$-$\overline{\mathrm{span}}\{T^n\}$.

Lemma 1.1.9. *Let $\mathcal{A}$ be a unital commutative Banach $*$-algebra such that for all $a \in \mathcal{A}$, $\|a^*a\| = \|a\|^2$. Then for self adjoint $a \in \mathcal{A}$, and $\tau \in sp(\mathcal{A})$, $\tau(a) \in \mathbb{R}$.*

Proof. Let $\tau(a) = \alpha + i\beta$ for $\alpha, \beta \in \mathbb{R}$. $\forall t \in \mathbb{R}$, we have

$$\begin{aligned}
|\tau(a) + it|^2 &= |\tau(a + it1)|^2 \\
&\leq \|a + it\|^2 \\
&= \|(a - it1)(a + it1)\| \\
&= \|a^2 + t^2 I\| \\
&\leq \|a^2\| + t^2 \\
&= \|a\|^2 + t^2
\end{aligned}$$

But $|\tau(a) + it|^2 = |\alpha|^2 + |\beta + t|^2 = |\alpha|^2 + |\beta|^2 + 2|t\beta| + |t|^2$ to begin with. But this gives

$$\alpha^2 + \beta^2 + 2|t\beta| \leq \|a\|^2$$

for all $t \in \mathbb{R}$, which is absurd unless $\beta = 0$. □

Theorem 1.1.4. *Let $\mathcal{A}$ be a unital commutative Banach $*$-algebra such that for all $a \in \mathcal{A}$, $\|a^*a\| = \|a\|^2$. Then the Gelfand transform is a bijective isometric $*$-homomorphism.*

Proof. We already have shown that $\Phi : \mathcal{A} \to C(\mathrm{sp}(\mathcal{A}))$ is a contractive algebra homomorphism.

And by Theorem 1.1.3 for every self adjoint $a \in \mathcal{A}$, $\|\Phi(a)\| = \|a\|$. As a consequence of Lemma 1.1.9, we get that $\Phi(a^*) = \overline{\Phi(a)}$. Then

verify how $\Phi(a^*) = \overline{\Phi(a)}$. This is evident if elements of $sp(\mathcal{A})$ preserve the involution

$$\begin{aligned}
\|\Phi(a)\|^2 &= \|\Phi(a)^*\Phi(a)\| \\
&= \|\Phi(a^*)\Phi(a)\| \\
&= \|\Phi(a^*a)\| \\
&= \|a^*a\| \quad \text{since it is self adjoint} \\
&= \|a\|^2
\end{aligned}$$

and we see that Φ is isometric.

It remains to show that Φ is surjective. First note that $\Phi(\mathcal{A})$ is a unital $*$-subalgebra of $C(\mathrm{sp}(\mathcal{A}))$. Thus by Stone-Weierstrass, we only need to show that $\Phi(\mathcal{A})$ separates points of $\mathrm{sp}(\mathcal{A})$. Let $\tau_1 \neq \tau_2 \in \mathrm{sp}(\mathcal{A})$. Then $\exists a \in \mathcal{A}$ such that $\Phi(a)(\tau_1) = \tau_1(a) \neq \tau_2(a) = \Phi(a)(\tau_2)$. Hence we are done. □

Corollary 1.1.4.1. *Let $T \in B(\mathcal{H})$ be a normal operator, and let $\mathcal{A} = \overline{span}\{T^n T^{*m} : m, n \in \mathbb{N} \cup \{0\}\}$. Then the Gelfand transform $\Phi : \mathcal{A} \to C(sp(\mathcal{A}))$ is halal (preserve all structure).*

Proof. Observe that $\mathcal{A}$ is a Banach * algebra with $\|T^*T\| = \|T\|^2$. Now we are left to prove that the C* identity hold in $\mathcal{A}$. verify. □

Lemma 1.1.10. *In the above corollary, the map $\psi : sp(\mathcal{A}) \to \sigma(T) := \tau \mapsto \tau(T)$ is a bijective homeomorphism.*

Proof. Lemma 1.1.8 shows that ψ is a well defined surjection. Assume $\psi(\tau) = \psi(\pi)$ for $\tau, \pi \in \mathrm{sp}(\mathcal{A})$. That is $\tau(T) = \pi(T)$. Then $\tau(T^m) = \pi(T^m)$ and $\tau(T^{*m}) = \pi(T^{*m})$ for all $m \in \mathbb{N} \cup \{0\}$. Moreover since τ, π are multiplicative by the definition, we see that τ, π agree on $\mathrm{span}\{T^n T^{*m} : m, n \in \mathbb{N} \cup \{0\}\}$. Thus $\tau = \pi$. Hence we see that ψ is a bijective map.

Now to prove the continuity, let τ_α be a net weak * converging to $\tau \in \mathrm{sp}(\mathcal{A})$. Then $\tau_\alpha(T) \to \tau(T)$ by definition. Thus we see that ψ is a continuous map. Now since $\mathrm{sp}(\mathcal{A})$ is compact as we proved in Lemma 1.1.4, and $\sigma(T) \subset \mathbb{C}$ is compact, by a general fact in topology, we get that ψ is a homeomorphism. □

1.2 Continuous Functional Calculus

Theorem 1.2.1 (Continuous functional calculus)**.** *Let $T \in B(\mathcal{H})$ be normal. Then $\exists$ a unique isometric $*$-homomorphism $\Xi : C(\sigma(T)) \to B(\mathcal{H})$ such that for any polynomial $p(z, \bar{z}) \in C(\sigma(T))$, we have*

$$\Xi(p(x)) = p(T, T^*)$$

For every $f \in C(\sigma(T))$, we denote $f(T) := \Xi(f)$ and call this the continuous functional calculus of T.

Proof. Existence of the map Ξ is guaranteed by Corollary 1.1.4.1 and Lemma 1.1.10. For the uniqueness note that the image of Ξ is the algebra in Corollary 1.1.4.1 and that the identity map $\sigma(T) \to \sigma(T)$ must be mapped to $T \in B(\mathcal{H})$. Since the algebra $\mathcal{A}$ is generated by T, the $*$-homomorphism ensure the uniqueness of Ξ. □

Definition 1.2.1. Let $(X, \mathcal{A})$ be a measurable space and $\mathcal{H}$ be a Hilbert space. A **spectral measure** on the triple $(X, \mathcal{A}, \mathcal{H})$ is a map $E : \mathcal{A} \to B(\mathcal{H})$ such that

(1) $E(A)$ is a projection, $\forall A \in \mathcal{A}$

(2) $E(A \cap B) = E(A)E(B), \forall A, B \in \mathcal{A}$

(3) $E(\emptyset) = 0, E(X) = I$

(4) If $\{A_n\}_{n\in\mathbb{N}}$ is a countable family of disjoint measurable sets, then

$$E\Big(\bigcup_{n\in\mathbb{N}} A_n\Big) = \sum_{n\in\mathbb{N}} E(A_n)$$

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Example 1.2.1. Let (X, Σ, μ) be a measure space. Define $E : \Sigma \to B(L^2(X, \mu)) := A \to M_{\chi_A}$. It is easy to see that E satisfies the first 3 properties of a spectral measure. To verify the fourth property, let $A_1, A_2, \ldots \in \Sigma$ be disjoint collection. Then

$$M_{\chi_{\cup_{n=1}^{\infty} A_n}}(f) = \chi_{\cup_{n=1}^{\infty} A_n} f = \sum_{n=1}^{\infty} \chi_{A_n} f = \sum_{n=1}^{\infty} M_{\chi_{A_n}} f$$

shows that the fourth property is also satisfied.

Proposition 1.2.1. *Let E be a spectral measure on $(X, \Sigma, \mathcal{H})$. Then for every $\xi, \eta \in \mathcal{H}$.*

$$E_{\xi,\eta}(A) = \langle E(A)\xi, \eta\rangle$$

defines a finite measure on (X, Σ) with $\|E_{\xi,\eta}\| \leq \|\xi\|\|\eta\|$.

Proof. That $E_{\xi,\eta}(\emptyset) = 0$ is evident. Hence we only need to verify the countable disjoint additivity to show that $E_{\xi,\eta}$ is a measure. Let $A_1, A_2, \ldots \in \Sigma$ be a mutually disjoint collection. Then

$$\begin{aligned}
E_{\xi,\eta}\Big(\bigcup_{n=1}^{\infty} A_n\Big) &= \Big\langle E\Big(\bigcup_{n=1}^{\infty} A_n\Big)\xi, \eta\Big\rangle \\
&= \Big\langle \sum_{n=1}^{\infty} E(A_n)\xi, \eta\Big\rangle \\
&= \sum_{n=1}^{\infty} \big\langle E(A_n)\xi, \eta\big\rangle \\
&= \sum_{n=1}^{\infty} E_{\xi,\eta}(A_n)
\end{aligned}$$

Taking the summation outside the inner product is justified by the below lemma. Thus we see that $E_{\xi,\eta}$ is a complex measure. Moreover, since $|\langle E(A)\xi, \eta\rangle| \leq \|\xi\|\|\eta\|$, as $E(A)$ is a projection, we see that the measure is finite (Although this is implicit in complex measures). To see that $\|E_{\xi,\eta}\| \leq \|\xi\|\|\eta\|$, use the definition of bounded variation and the lemma below. $\square$

Lemma 1.2.1. *Let $\{P_i\}_{i\in I}$ be a family of pairwise orthogonal projections on a Hilbert space $\mathcal{H}$. Then there exist a unique projection P on $\mathcal{H}$ such that $\forall \xi, \eta \in \mathcal{H}$,*

$$\sum_{i\in I} \langle P_i\xi, \eta\rangle = \langle P\xi, \eta\rangle$$

This means the convergence of orthogonal projections is not in the norm sense, but rather in the sense above. For example

Example 1.2.2. Consider $P_n = P_{\delta_n}$ in $B(\ell^2(\mathbb{N}))$, and let $Q_n = \sum_{i=1}^n P_i$. Then clearly Q_i doesn't converge in norm, but rather in the above sense to the identity map in $B(\ell^2(\mathbb{N}))$.

I'm not convinced. Need to verify

Lemma 1.2.2 (Reisz representation for sesquilinear forms)**.** *Let $B(\cdot,\cdot)$ be a bounded sesquilinear form on a Hilbert space $\mathcal{H}$. Then there exist a unique $T \in B(\mathcal{H})$ such that for all $\xi, \eta \in \mathcal{H}$,*

$$B(\xi, \eta) = \langle T\xi, \eta\rangle$$

verify if ϕ is a linear functional

Proposition 1.2.2. *Let E be a spectral measure on $(X, \Sigma, \mathcal{H})$, and ϕ be a bounded linear functional on X. Then there exist a unique $T_\phi \in B(\mathcal{H})$ such that for all $\xi, \eta \in \mathcal{H}$*

$$\int_X \phi \, dE_{\xi,\eta} = \langle T_\phi\xi, \eta\rangle$$

and we denote $T_\phi := \int_X \phi \, dE$

Proof. See that the integral is sesquilinear on ξ, η for simple functions. Then use Lemma 1.2.2. □

Note that the set $\mathcal{M}(X)$ of all bounded measurable functions on X, equipped with sup norm is a Banach space. Moreover with pointwise product and complex conjugation, it turns into a commutative C* algebra.

verify if ϕ is a linear functional

Theorem 1.2.2. *Let E be a spectral measure on $(X, \Sigma, \mathcal{H})$ and ϕ be a bounded measurable function. The map*

$$\mathcal{M}(X) \to B(\mathcal{H}) := \phi \mapsto \int_X \phi \, dE$$

is a contractive $*$-homomorphism.

Proof. To show that it is a $*$-homomorphism, observe that $\overline{E_{\xi,\eta}} = E_{\eta,\xi}$. This follows since

$$\overline{E_{\xi,\eta}(A)} = \overline{\langle E(A)\xi, \eta\rangle} = \langle \eta, E(A)\xi\rangle = \langle E(A)\eta, \xi\rangle = E_{\eta,\xi}(A)$$

Now let $T_{\overline{\phi}} \in B(\mathcal{H})$ corresponding to ϕ as in Proposition 1.2.2. Then by definition,

$$\int_X \overline{\phi}\ dE_{\xi,\eta} = \langle T_{\overline{\phi}}\eta, \psi\rangle$$

Now the left integral is equal to

$$\overline{\int_X \phi\ dE_{\eta,\xi}} = \overline{\langle T_\phi \eta, \xi\rangle} = \langle \xi, T_\phi \eta\rangle = \langle T_\phi^* \xi, \eta\rangle$$

Thus we get that $T_{\overline{\phi}} = T_\phi^*$ and hence the map preserve the involution.

To show that the map is multiplicative, we need to show that

$$\int_X \phi\psi\ dE = \int_X \phi\ dE \circ \int_X \psi\ dE$$

Notice that by Proposition 1.2.2, we'll done if we show that for all $\xi, \eta \in \mathcal{H}$,

$$\int_X \phi\psi\ dE_{\xi,\eta} = \int_X \phi\ dE_{(\int_X \psi\ dE)\xi,\eta}$$

But for this, it is enough to show the equivalence of the measures $\psi dE_{\xi,\eta}$ and $E_{(\int_X \psi\ dE)\xi,\eta}$. That is for all $A \in \Sigma$, we need to show that

$$\Big\langle E(A)\Big(\int_X \psi\ dE\Big)\xi, \eta\Big\rangle = \int \psi\chi_A\ dE_{\xi,\eta} \tag{1.1}$$

But again, the left hand side of the inner product is

$$\Big\langle \Big(\int_X \psi\ dE\Big)\xi, E(A)\eta\Big\rangle$$

since $E(A)$ is a projection. Again by Proposition 1.2.2, we see that the above inner product is

$$\int_X \psi\ dE_{\xi,E(A)\eta}$$

Then Equation 1.1 reduces to showing

$$\int_X \psi\ dE_{\xi,E(A)\eta} = \int \psi\chi_A\ dE_{\xi,\eta}$$

Again using the same reasoning, it is enough to show that the measures are the same. That is for any $B \in \Sigma$, we must have

$$E_{\xi,E(A)\eta}(B) = \int_X \chi_A \chi_B \; dE_{\xi,\eta}$$

But this is equivalent to

$$\langle E(A)\xi, E(B)\eta\rangle = \langle E(A)E(B)\xi, \eta\rangle = \langle E(A\cap B)\xi, \eta\rangle = \int_X \chi_{A\cap B} \; dE_{\xi,\eta} = E_{\xi,\eta}(A\cap B)$$

which is true by a property of the Spectral measure. $\square$

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Theorem 1.2.3. *Let X be a compact Hausdorff space and $\mathcal{H}$ be a Hilbert space, and $\Psi : C(X) \to B(\mathcal{H})$ is a $*$-homomorphism (representation). Then there exist a unique spectral measure on $(X, \mathcal{B}, \mathcal{H})$ ($\mathcal{B}$ being the Borel sigma algebra on X), such that*

$$\Psi(f) = \int_X f \; dE$$

Proof. For every $\xi, \eta \in \mathcal{H}$, the map $\nu : C(X) \to \mathbb{C} := f \mapsto \langle \Psi(f)\xi, \eta\rangle$ is a bounded linear functional on $C(X)$, hence by the reisz representation theorem, there exists a unique measure $\mu_{\xi,\eta}$ on X such that

$$\langle \Psi(f)\xi, \eta\rangle = \int f \; d\mu_{\xi,\eta} \quad f \in C(X)$$

Now for any $\phi \in \mathcal{M}(X)$, the map

$$\mathcal{H} \times \mathcal{H} \to \mathbb{C} := (\xi, \eta) \mapsto \int f \; d\mu_{\xi,\eta}$$

is a bounded sesquilinear form with norm $\|\xi\|\|\eta\|$. Hence there exist a unique $T_\phi \in B(\mathcal{H})$ such that

$$\langle T_\phi \xi, \eta\rangle = \int \phi \; d\mu_{\xi,\eta}$$

Now we show that the map $\mathscr{F} : \mathcal{M}(X) \to B(\mathcal{H}) := \phi \mapsto T_\phi$ is a $*$-representation of $\mathcal{M}(X)$. Linearity is obvious, though verify. Argument for $T_{\overline{\phi}} = T_\phi^*$ is as in the last theorem. Observe that for every $f \in C(X)$, $T_f = \Psi(f)$. Take inner product with ξ, η.

To show multiplicativity of the above map, let $f \in C(X), \phi \in \mathcal{M}(X)$. Then $\forall \xi, \eta \in \mathcal{H}$,

$$\begin{aligned}\langle T_{\phi f}(\xi), \eta\rangle &= \int \phi f \; d\mu_{\xi,\eta} \\ &= \int \phi f \; d\mu_{\Phi(f)\xi,\eta} \\ &= \langle T_\phi \Phi(f)\xi, \eta\rangle\end{aligned}$$

shows that $\mathscr{F}$ is multiplicative if one of the functions are in $C(X)$. More generally, if $\phi, \psi \in \mathcal{M}(X)$, choose a net $f_i \in C(X)$ such that f_i converges to ϕ weak * in $C(X)^{**}$, and $\|f_i\| \to \|\phi\|$ (This is guaranteed by Goldstein's theorem) and $\|f_i\| \leq \|\phi\|$. Now for any $\xi, \eta \in \mathcal{H}$, we have

$$\langle T_{\phi,\psi}\xi, \eta\rangle = \int \phi\psi \; d\mu_{\xi,\eta} = \lim_i \int \phi f_i \; d\mu_{\xi,\eta} = \lim_i \langle T_{\phi f_i}\xi, \eta\rangle = \lim_i \langle T_\phi T_{f_i}\xi, \eta\rangle$$

which converge to $\langle T_\phi T_\psi \xi, \eta\rangle$. verify. Thus $\mathscr{F}$ is a $*$-representation of $\mathcal{M}(X)$.

Now define $E : \Sigma(A) \to B(\mathcal{H}) := A \mapsto T_{\chi_A}$. Obviously E satisfy all the first three properties of a spectral measure easily.

See also that the map E is countably additive. Let (A_n) be a family of pairwise disjoint measurable sets of X and let $A = \cup_{n=1}^\infty A_n$. Then for every $\xi, \eta \in \mathcal{H}$,

$$\Big\langle E\big(\bigcup_{n\in\mathbb{N}} A_n\big)\xi, \eta\Big\rangle = \int \chi_{\cup_{n=1}^\infty A_n} \; d\mu_{\xi,\eta} = \mu_{\xi,\eta}\Big(\bigcup_{n=1}^\infty A_n\Big) = \sum_{n\in\mathbb{N}} \mu_{\xi,\eta}(A_n)$$

□

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Theorem 1.2.4 (Spectral decomposition)**.** *Let $T \in B(\mathcal{H})$ be a normal operator. Then there exists a unique spectral measure $E \in (\sigma(T), \Sigma, \mathcal{H})$ such that*

$$T = \int_{\sigma(T)} \lambda \; dE(\lambda)$$

Proof. Immediate from what we've proven so far?? verify. . □

See $\sigma(T)$ as $sp(T)$ also

Let X be a compact subset of $\mathbb{C}$. Let μ be a Radon measure on X. Let $\mathcal{H} = L^2(X, \mu)$, and $T \in B(L^2(X, \mu))$ by defining $(T\eta)(x) = x\eta(x)$ almost everywhere. Then $(T^*\eta)(x) = \bar{x}\eta(x)$. Then notice that T is normal. Define spectral measure by $E(A) = M_{\xi_A}$.

Definition 1.2.2. Let $\mathcal{A}$ be a $*$-subalgebra of $B(\mathcal{H})$. A vector $\eta \in \mathcal{H}$ is called cyclic for $\mathcal{A}$ iff $\{a\eta \; : \; a \in \mathcal{A}\}$ is dense in $\mathcal{H}$.

Definition 1.2.3. Let $T \in B(\mathcal{H})$ be normal. We denote $C^*(T) = \overline{\text{span}}\{T^n T^{*m} : m, n \geq 0\}$

Let $\mathcal{A}$ be a *-subalgebra of $B(\mathcal{H})$, and $\eta \in \mathcal{H}$. Define $\langle \cdot, \cdot \rangle : \mathcal{A} \times \mathcal{A} \to \mathbb{C} := \langle a, b \rangle' := \langle a\eta, b\eta \rangle_{\mathcal{H}}$

Lemma 1.2.3. *Let $\mathcal{A}$ be a $*$-subalgebra of $B(\mathcal{H})$, and let $\eta \in \mathcal{H}$. Then η is cyclic for $\mathcal{A}$ if and only if it is separating for $\mathcal{A}'$, the commutant of $\mathcal{A}$.*

Proof. verify □

If $\mathcal{A}$ is commutative and $\eta \in \mathcal{A}$ is cyclic for $\mathcal{A}$, then $\langle \cdot, \cdot \rangle$ is an inner product on $\mathcal{A}$.

Denote $\Xi : \mathcal{A} \to \mathcal{H} := a \mapsto a\eta$. Then note that this map is an isometry under the new inner product.

Theorem 1.2.5. *Let $T \in B(\mathcal{H})$ be normal. Assume there exists a cyclic vector $\eta \in \mathcal{H}$ for $C^*(T)$. Then there exists a Radon probability measure $\mu \in Prob(sp(T))$ and a unitary*

$$U : \mathcal{H} \to L^2(sp(T), \mu)$$

such that $UTU^* = M_I$, where I is the identity function on $sp(T)$.

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Lemma 1.2.4. *Let μ, μ' be Radon probability measures on X, such that $\mu \sim \mu'$. Define $U : L^2(X, \mu) \to L^2(X, \mu')$ by sending*

$$U : \xi \to \sqrt{\frac{d\mu}{d\mu'}}\xi$$

Then U is a unitary, and $\forall f \in C(X)$, and

$$UM_f U^* = M'_f$$

where $M_f \in B(L^2(X, \mu))$ and $M'_f \in B(L^2(X, \mu'))$.

Proof. To see that U is an isometry, take the L^2 norms on both of these spaces. verify the rest. □

Lemma 1.2.5. *Let $\mu, \mu' \in Prob(X)$. If $\exists$ unitary $U : L^2(\mu) \to L^2(\mu')$ such that $UM_f U^* = M_f$ for all $f \in C(X)$, then $\mu \sim \mu'$.*

Proof. Let $h = |U1|^2 \in L^1(\mu')$. $\forall f \in C(X)$,

$$\int f \, d\mu = \langle M_f 1, 1\rangle_\mu = \langle U^* M_f' U 1, 1\rangle = \int fh \, d\mu'$$

Thus we get $\mu = h\mu'$. Hence we see that $\mu \sim \mu'$. □

Definition 1.2.4 (Strong operator topology)**.** The strong operator topology is the locally convex topology on $B(\mathcal{H})$ for separable $\mathcal{H}$, defined by the family of seminorms $\{\phi_x \ : \ x \in \mathcal{H}\}$, where $\phi_x(T) := \|Tx\|$. We have $T_i \overset{SOT}{\to} T$ if and only if $\|(T_i - T)x\| \to 0$ for all $x \in \mathcal{H}$. For any $x_1, x_2, \ldots, x_n \in \mathcal{H}$, and $\varepsilon > 0$, $T \in B(\mathcal{H})$, the set

$$\{S \in B(\mathcal{H}) \ : \ \|(T - S)x_i\| < \varepsilon, \ \forall i = 1, 2, \ldots n\}$$

is an open neighborhood of T.

Definition 1.2.5. The weak operator topology is the locally convex topology on $B(\mathcal{H})$ for separable $\mathcal{H}$, defined by the family of seminorms $\{\phi_{x,y} \ : \ x \in \mathcal{H}\}$, where $\phi_{x,y}(T) := |\langle Tx, y\rangle|$. We have $T_i \overset{WOT}{\to} T$ if and only if $|\langle (T_i - T)x, y\rangle| \to 0$ for all $x, y \in \mathcal{H}$. For any $x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_n \in \mathcal{H}$, and $\varepsilon > 0$, $T \in B(\mathcal{H})$, the set

$T_i \to T$ in WOT
$T_i(x) \overset{w^*}{\to} T(x)$ for all $x \in \mathcal{H}$

$$\{S \in B(\mathcal{H}) \ : \ |\langle (T - S)x_i, y_i\rangle| < \varepsilon, \ \forall i = 1, 2, \ldots n\}$$

is an open neighborhood of T.

Example 1.2.3. Let $\{f_i\} \subset C([0,1]) \ni f$ identified as multiplication operators on $B(L^2([0,1]))$. Then $M_{f_i} \overset{WOT}{\to}$ if and only if $f_i \to f$ in weak * topology on $L^\infty([0,1])$.

Proof. verify with intergrals and the fact with dual space of $L^1([0,1])$ □

Lemma 1.2.6. *For any convex subset $K \subset B(\mathcal{H}), \overline{K}^{SOT} = \overline{K}^{WOT}$.*

Proof. Clearly $\overline{K}^{\text{SOT}} \subset \overline{K}^{\text{WOT}}$ since the topologies are relative. Let $T \in \overline{K}^{\text{WOT}}$, $x_1, x_2, \ldots, x_n \in H$ and $\varepsilon > 0$. Let $\mathcal{H}^n = \mathcal{H} \oplus \mathcal{H} \oplus \ldots \oplus \mathcal{H}$, and $T^n \in B(H^n)$ be defined as $T^n = T \otimes C_n$, then n-amplification of T. Similarly, let

$$K^n = \{T^n \ : \ T \in K\}.$$

Then K^n is a convex subset of $B(\mathcal{H}^n)$, and

$$K^n(x_1, x_2, \ldots, x_n) = \{T^n(x_1, x_2, \ldots, x_n) \ : \ T \in K\}$$

is a convex subset of $\mathcal{H}^n$. Thus, the weak and norm closures of the set coincide by what we proved last semester. By assuming, $T^n(x_1, x_2, \ldots, x_n)$ in the weak closure of $K^n(x_1, x_2, \ldots, x_n)$, there is a $T_i \in K$ such that $T_i^n(x_1, x_2, \ldots, x_n) \to T^n(x_1, x_2, \ldots, x_n)$ weakly. □

Exercise 1.2.1. Explore the correspondance between states on $\ell^\infty(X)$ and the finitely additive probability measures on X.

Theorem 1.2.6. *Let $\mathcal{H}$ be a Hilbert space, and let $f : B(\mathcal{H}) \to \mathbb{C}$ be linear. Then, the following are equivalent.*

(1) f is SOT continuous.

(2) f is WOT continuous.

(3) There exists $\xi_1, \xi_2, \ldots, \xi_n, \eta_1, \eta_2, \ldots, \eta_n \in \mathcal{H}$ such that

$$f(T) = \sum_{i=1}^{n} \langle T\xi_i, \eta_i \rangle$$

Proof. We just need to prove 1 $\implies$ 3. By a lemma from last semester, there must exist $\xi_1, \xi_2, \ldots, \xi_n \in \mathcal{H}$ such that $|f(T)| \leq \sum_{i=1}^{n} \|T\xi_i\|$ for all $T \in B(\mathcal{H})$. Let $\mathcal{K} = \overline{\{T\xi_o \ : \ T \in B(\mathcal{H})\}} \subset \mathcal{H}$. Then

$$\phi : \mathcal{K} \to \mathbb{C} := T\xi_0 \to f(T)$$

is a well defined linear functional for $\mathcal{K}$. Now by Reisz representation, we see that $\phi(T\xi_0) = \langle T\xi_0, \eta_0 \rangle$ for some $\eta_0 \in \mathcal{H}$.

But here, for $\xi_1, \xi_2, \ldots, \xi_n$, consider them in $\mathcal{H}^n$ and do the same process to get a $(\eta_1, \eta_2, \ldots, \eta_n) \in \mathcal{H}^n$. $\square$

Theorem 1.2.7. *Let X, Y be normed spaces. For $1 \leq p \leq \infty$, define*

$$\|(x, y)\|_p = \Big(\|x\|^p + \|y\|^p \Big)^{\frac{1}{p}}$$

Then $\|\cdot\|_p$ is a norm extending both norms in X and Y. If X and Y are Banach, then so is $(X \oplus Y, \|\cdot\|_p)$. All these norms are equivalent for all $1 \leq p \leq \infty$.

Proposition 1.2.3. *We have that $(X \oplus_p Y)^* = X^* \oplus_q Y^*$.*

Proof. We first show when $p = 1, q = \infty$. Consider the map

$$\Xi : (X \oplus_1 Y)^* \to X^* \oplus_\infty Y^* := F \to F_X \oplus F_Y$$

once we identify X as $X \oplus 0$ and Y similarly as subspaces of $X \oplus Y$. See that this is a bijective linear map. Now show that the norms are preserved in Ξ. $\square$